Gaussian Splatting and NeRF
Supplementary tutorial

Computer Vision 2026
University of Bern · Computer Vision Group

Interactive lab: sosekie.github.io/Unibe-CVG-CV-Interactive-Labs/gaussian-splatting-nerf/
Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Learning objectives

- Connect Gaussian shape to covariance eigenvectors, variances, and normalization.
- Distinguish coefficient fitting, geometry fitting, and image-based inverse rendering.
- Explain projection, visibility, optical thickness, and pixel gradients.
- Compare Gaussian primitives with neural fields and design controlled evaluation.

Key concepts: identifiability, conditioning, covariance, gradient flow, compositing, multi-view geometry, density control, ray integration, and generalization.

Section	Experiment	Questions
1	Approximate a function in 1D	1.1 and 1.2
2	Covariance and a 2D input domain	2.1 and 2.2
3	Fit a 3D scalar function	3.1 and 3.2
4	Project, composite, and expose ambiguity	4.1 and 4.2
5	Recover a scene from images	5.1 and 5.2
6	NeRF: a neural field along camera rays	6.1 and 6.2
7	Transfer to the real captured Garden	7.1 and 7.2

Working with the tutorial

Make a prediction, try each guided experiment, and record observations. Choose an answer to each question and justify it; the explanation and follow-up matter more than the letter. Consult the separate solutions after your attempt.

This handout follows the reviewed interactive tutorial. Lecture-slide references and a numbered course worksheet assignment remain pending. The NeRF experiment trains a small 2D cross-section; the final Garden renders a pretrained Gaussian Splatting model.

> Local use: serve the lab's **dist/** directory with Python 3. Sections 1–6 use no remote assets. The optional Garden requires network access and WebGL and streams approximately 187 MB.

References: Kerbl et al., 3D Gaussian Splatting (2023); Mildenhall et al., NeRF (2020). See the lab README for conventions and teaching limits.

1. Approximate a function in 1D

Open this experiment in the interactive lab ↗

Learning objective: Distinguish amplitude fitting, basis geometry, and identifiability.

Guided experiment

1. Fit the smooth-bump target with N = 2 and N = 8. Record sample RMSE and basis rank.
2. Make two kernels identical in mean and width. Predict the rank and possible amplitude solutions before fitting.
3. Try the signed-oscillation target. Inspect the individual contributions and explain negative coefficients.

Observations: Record settings, numerical results, visible effects, and your explanation.

1.1 · IDENTIFIABILITY · LINEAR FITTING

You fit two 1D Gaussian basis functions with exactly the same mean and width. Only their amplitudes are free. The target samples are fixed.

What can least squares determine about the two amplitudes?

A. A unique pair, provided there are more samples than unknown amplitudes.

B. Their sum can be determined, but the individual amplitudes are not identifiable.

C. Both amplitudes must be equal because the two kernels have the same shape.

Explain further: Would adding a penalty $\lambda\|a\|^2$ make the amplitudes identifiable from the observations?

1.2 · MODEL ASSUMPTIONS · SIGNED WEIGHTS

After an unconstrained amplitude fit, a nonnegative target is approximated using both positive and negative Gaussian coefficients. Small regions of the approximation fall below zero.

Which interpretation is consistent with the model used in this lab?

A. The negative coefficients are invalid because every Gaussian expansion is a probability mixture.

B. The negative coefficients show that least squares has failed to reach its optimum.

C. Signed coefficients are valid for function approximation; nonnegativity would require a different constrained fit.

Explain further: Why can a low sample error coexist with large, unstable positive and negative coefficients?

2. Covariance and a 2D input domain

Open this experiment in the interactive lab ↗

Learning objective: Read Gaussian geometry from covariance eigenpairs and distinguish scale from normalization.

Guided experiment

1. Set one Gaussian's scales to 0.60 and 0.20 and its rotation to 45°. Inspect its covariance.
2. Make it isotropic and rotate again. Explain which quantities stop changing and why.
3. Isolate one contribution and double both scales at fixed amplitude. Predict the peak and full-plane integral; account for the finite plot and automatic display scaling.

Observations: Record settings, numerical results, visible effects, and your explanation.

2.1 · COVARIANCE · PRINCIPAL AXES

A 2D Gaussian has $\Sigma = [[0.20, 0.16], [0.16, 0.20]]$. The diagonal entries are equal.

What shape does this covariance describe?

A. An elongated Gaussian, with standard deviations 0.6 and 0.2 along axes rotated by 45°.

B. An isotropic Gaussian with standard deviation $\sqrt{0.20}$ in every direction.

C. An elongated Gaussian aligned with the x and y axes, with scales 0.20 and 0.16.

Explain further: What happens to the usefulness of a rotation gradient as the two principal scales become equal?

2.2 · GAUSSIAN BEHAVIOR · NORMALIZATION

Consider one unnormalized 2D kernel $a \exp(-\tfrac{1}{2}(x-\mu)^{\top}\Sigma^{-1}(x-\mu))$. Double both principal scales while keeping its amplitude a fixed.

What happens to its peak value and its integral over the whole plane?

A. The peak becomes one quarter as large and the integral stays fixed.

B. The peak stays fixed and the integral becomes four times as large.

C. The peak stays fixed and the integral becomes twice as large.

Explain further: Does the fourfold integral increase imply fourfold image brightness after projecting and alpha-compositing a 3D splat?

3. Fit a 3D scalar function

Open this experiment in the interactive lab ↗

Learning objective: Separate approximation capacity, initialization, and optimization error.

Guided experiment

1. Choose the Gaussian-bump target and N = 1. Set amplitude to zero, record geometry, and apply one update. Explain the raw-gradient effect.
2. Fit the ring target with N = 1 and N = 8 using comparable update budgets. Inspect all three residual slices.
3. Rotate the inspection view without editing the fit. Explain why the sampled scalar objective stays fixed.

Observations: Record settings, numerical results, visible effects, and your explanation.

3.1 · OPTIMIZATION · GRADIENT FLOW

The 3D model is $\hat{y}(x) = a\varphi(x;\mu,\Sigma)$, fitted by mean squared error to sampled target values. Consider the raw gradient at $a = 0$, with positive scales.

Which parameters can receive a nonzero loss gradient at that instant?

A. The mean and covariance can receive nonzero gradients, while amplitude cannot.

B. All parameter gradients must be zero because the prediction is zero.

C. Amplitude can receive a nonzero gradient, while the mean and covariance gradients are zero.

Explain further: Can Adam momentum move a mean when its current raw gradient is zero?

3.2 · APPROXIMATION · CAPACITY VS OPTIMIZATION

One Gaussian fits the ring-shaped 3D target poorly after many updates. Its slices fill the hole or miss much of the ring.

What can be concluded from the Gaussian model itself, without assuming the optimizer found a global minimum?

A. One Gaussian cannot represent the target exactly; more optimization may help, but cannot remove this representation mismatch.

B. The remaining error proves that Adam is stuck at a local minimum.

C. A sufficiently small learning rate guarantees that one Gaussian will reproduce the ring.

Explain further: If the displayed volume RMSE is small, could a visually important part of the ring still be poorly fitted?

4. Project, composite, and expose ambiguity

Open this experiment in the interactive lab ↗

Learning objective: Relate visibility to gradients and distinguish rotation from translation.

Guided experiment

1. Use three-splat overlap. Sample one pixel, vary foreground opacity, and follow the rear colour weights.
2. Keep one Gaussian. Record its projected axes, move it along the viewing ray with the provided scale change, and compare.
3. Change camera yaw to expose the 3D difference. Here yaw orbits the camera centre; explain why pure rotation at a fixed centre is different.

Observations: Record settings, numerical results, visible effects, and your explanation.

4.1 · VISIBILITY · PHOTOMETRIC GRADIENTS

At one pixel, a foreground splat has $\alpha_1 = 0.98$ and a rear splat has $\alpha_2 = 0.6$. Hold geometry and opacity fixed while differentiating the pixel colour with respect to the rear RGB.

What is $\partial C/\partial c_2$ for each RGB channel, and what does it imply for fitting?

A. 0.6: the rear splat receives its full opacity weight regardless of foreground visibility.

B. 0.012: foreground transmittance strongly attenuates the rear colour gradient at this pixel.

C. 0: any foreground alpha greater than zero completely blocks the rear colour gradient.

Explain further: Would directly adding the two coloured kernels give the same learning behavior?

4.2 · MULTI-VIEW GEOMETRY · AMBIGUITY

For an isolated Gaussian and a fixed pinhole camera, multiply its camera-relative mean and all spatial scales by $k > 0$. Keep RGB and opacity fixed, and keep it in front of the near plane. Its locally projected footprint is unchanged.

Which extra observation generally constrains this depth–scale ambiguity?

A. A second image obtained by rotating the camera about the same optical centre.

B. The same camera image rendered at a higher pixel resolution.

C. A calibrated image from a different camera centre, with parallax and visibility of the Gaussian.

Explain further: Why can three nearby training views provide less useful depth information than three spread-out views?

5. Recover a scene from images

Open this experiment in the interactive lab ↗

Learning objective: Design controlled comparisons and distinguish image agreement from geometric evidence.

Guided experiment

1. Fit one training view with 24 Gaussians for 100 updates. Record training and reserved-view MSE.
2. Restart with three spread-out views, using the same budget and update count. Compare both errors and residual structure.
3. Record error and count, apply density refinement once, then fit 25 further updates. Separate the immediate model change from subsequent learning.

Observations: Record settings, numerical results, visible effects, and your explanation.

5.1 · EXPERIMENT DESIGN · GENERALIZATION

A fit from one selected camera has low training MSE but a much larger reserved-view MSE. You want to test whether limited view coverage is contributing to the failure.

Which comparison provides the most relevant controlled evidence?

A. Fit one view and three spread-out views from the same seeds, Gaussian budget, shape mode, and update budget; evaluate both at the reserved pose.

B. Give the one-view run many more Gaussians and a longer update budget, then compare its training error with the previous reserved error.

C. Use the reserved image as another training view and report its lower error as evidence of generalization.

Explain further: If you repeatedly choose budgets and stop times using frame 10, is it still an untouched final test?

5.2 · DENSITY CONTROL · MODEL CHANGES

Consider one isolated coloured splat with opacity o. Replace it by two coincident splats with identical shape and colour, each with opacity $o' = 1 - \sqrt{(1-o)}$. Let φ be their shared projected Gaussian value at a pixel.

Does this opacity adjustment preserve the rendered image everywhere?

A. Yes: matching opacity at the centre preserves the whole Gaussian footprint.

B. Yes: two coincident Gaussian contributions add linearly under alpha compositing.

C. No: it matches alpha at $\varphi = 1$, but the composited falloff generally differs elsewhere.

Explain further: For an additive scalar field, would two coincident copies with amplitudes $a/2$ preserve the function?

6. NeRF: a neural field along camera rays

Open this experiment in the interactive lab ↗

Learning objective: Distinguish density, segment opacity, quadrature, and neural-field capacity.

Guided experiment

1. At yaw 90° and the centre ray, compare density with termination weights. Explain the weak contribution of hidden orange density.
2. Compare 8 and 64 ray samples with fixed network weights. Track interval length, alpha, and total transmittance.
3. Fit 50 updates and inspect the reserved 0° view. Compare errors and explain why additional ray samples do not add network parameters.

Observations: Record settings, numerical results, visible effects, and your explanation.

6.1 · VOLUME RENDERING · OPTICAL THICKNESS

A ray crosses a homogeneous slab of length $L = 1$ with density $\sigma = 2$ and constant RGB. You approximate it with N equal intervals of length $\delta = L/N$, using $\alpha_i = 1 - \exp(-\sigma\delta)$.

What happens to total transmittance if you change N from 10 to 100?

A. It decreases by a factor of ten because the ray encounters ten times as many samples.

B. It stays $\exp(-2)$; each smaller interval has proportionally less optical thickness.

C. It becomes zero because enough samples make any positive density fully opaque.

Explain further: If a thin high-density feature lies entirely between the sample points, does a low sampled density prove the ray is empty?

6.2 · NEURAL FIELDS · MULTI-VIEW CONSISTENCY

In a classical NeRF-style model, density is $\sigma\theta(x)$, while RGB is $c\theta(x,d)$. A proposed variant also lets density depend freely on viewing direction: $\sigma\theta(x,d)$.

Why is position-only density a useful constraint for reconstructing a static scene?

A. It makes all colour independent of camera direction and therefore guarantees Lambertian appearance.

B. It guarantees unique geometry whenever training pixel error is small.

C. It makes different cameras query the same spatial extinction field, while allowing directional appearance in the RGB branch.

Explain further: Is increasing the number of samples per ray the NeRF equivalent of adding more Gaussian primitives?

7. Transfer to the real captured Garden

Open this experiment in the interactive lab ↗

Learning objective: Evaluate appearance and geometry claims at scene scale.

Guided experiment

1. Load the Garden when network and WebGL are available. It streams approximately 187 MB and renders Gaussian Splatting.
2. Inspect thin structures and viewpoints away from the comfortable viewing region. Record two artifacts and competing explanations.
3. Relate observations to coverage, representation, or appearance; identify evidence needed before using this renderer as collision geometry.

Observations: Record settings, numerical results, visible effects, and your explanation.

7.1 · APPEARANCE · REPRESENTATION CHOICE

Suppose a glossy leaf changes brightness with viewing direction in the captured photographs, but a reconstruction represents each Gaussian with one fixed RGB value. Even with good spatial coverage, a highlight appears painted onto the surface.

Which model change directly addresses this missing behavior?

A. Increase the image resolution while keeping the same trained constant-RGB splats.

B. Let each Gaussian's emitted colour depend on viewing direction, for example through spherical harmonics.

C. Normalize every spatial Gaussian to integrate to one.

Explain further: Could a low photometric error prove that positions and covariances are correct when colour is very flexible?

7.2 · TRANSFER · RENDERING VS GEOMETRY

The garden looks sharp from several viewpoints. A colleague wants to treat the visible Gaussian ellipsoids as a reliable surface model for collision detection.

What does the rendering evidence justify?

A. It supports appearance quality from those views; reliable collision geometry needs additional geometric validation and a defined surface representation.

B. Sharp novel views establish that every 1σ ellipsoid lies on the true surface.

C. Enough high-opacity Gaussians guarantee a watertight boundary, even in unobserved regions.

Explain further: If rendering at twice the resolution reveals smaller floaters, has the underlying geometry become worse?